



Task 3: Model Validation & Hyperparameter Tuning Report

1. Overfitting Analysis

During initial evaluations, an unconstrained Decision Tree Regressor was implemented. The baseline model achieved a near-zero training RMSE, indicating perfect memorization of the training dataset's noise and variance. However, when tested against unseen validation splits, the error expanded dramatically. This significant performance gap confirmed high variance and structural overfitting, proving that an unconstrained tree generalizes poorly to real-world deployment data.

2. Hyperparameter Tuning Approach

To counter the overfitting behavior, a structured optimization strategy was established using GridSearchCV paired with 5-Fold Cross-Validation. The parameter grid systematically tested structural depth constraints (`max_depth`) and minimum splitting thresholds (`min_samples_split`) across 5 distinct data folds to reduce reliance on a single train-test split. This automated process safely restricted the tree from making over-complex, isolated leaf splits.

3. Experimental Results

The final metrics across the evaluation pipeline are summarized below:

- **Linear Regression (Baseline):** RMSE: 0.7455 | R^2 Score: 57.58%
- **Ridge Regression (L2 Regularized):** RMSE: 0.7455 | R^2 Score: 57.58%
- **Tuned Decision Tree (Optimized):** RMSE: **0.6346** | R^2 Score: **69.26%**

4. Model Selection Justification

The **Tuned Decision Tree** was selected as the final production-ready model. By restricting unchecked depth variance, the tuned tree successfully minimized validation errors compared to the overfit model. It statistically outpaced the baseline linear variants by capturing non-linear interactions and regional geographic clusters embedded within the California dataset—patterns that strict mathematical linearity constraints cannot map. The stability of these metrics across 5-fold cross-validation confirms that the model is robust and ready for production deployment.